

Checkpoint 1 — Normal Equations

INFO 521 — Machine Learning Foundations

i Course-schedule placement

Week 2 · after **Module 1 — Linear Models & Least Squares** · gates **Project 1.1**. (Syllabus §1.6: “Checkpoint 1 (proposed)”, Wk 2.)

Gates: Project 1.1 (Least Squares) **Format:** in-class, closed-book, ~15 min, one derivation. Graded Satisfactory / Not-Yet. **Retakeable** (an alternate version is provided below).

Prompt (Version A)

For the single-predictor linear model $f(x; w_0, w_1) = w_0 + w_1 x$ with squared-error loss

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N (y_n - (w_0 + w_1 x_n))^2,$$

derive the least-squares estimator $\hat{w}_0$ (you may leave it in terms of $\hat{w}_1$). Your final answer must contain no summation symbols; use $\bar{a} = \frac{1}{N} \sum_n a_n$.

Prompt (Version B — retake)

Write the squared-error loss for a general linear model in matrix/vector form and derive the normal equations $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ by setting the gradient to zero. State the one assumption needed to invert.

Satisfactory criteria

- Sets $\partial \mathcal{L} / \partial w_0 = 0$ (A) or $\nabla_{\mathbf{w}} \mathcal{L} = \mathbf{0}$ (B).
- Algebra is correct and complete to the stated end result.
- (B) States that $\mathbf{X}^\top \mathbf{X}$ must be invertible (full column rank).

Answer key

A: $\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$.

B: From $\mathcal{L} = \frac{1}{N} (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w})$, $\nabla_{\mathbf{w}} \mathcal{L} = -\frac{2}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) = \mathbf{0} \Rightarrow \mathbf{X}^\top \mathbf{X} \hat{\mathbf{w}} = \mathbf{X}^\top \mathbf{y} \Rightarrow \hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, valid when $\mathbf{X}^\top \mathbf{X}$ is invertible.